The Apple Extract

Kaggle is a data science website listing data sets for investigation. I am working with their apple quality data set. Potential significant factors that I will evaluate are weight, size, sweetness, crunchiness, juiciness, ripeness, acidity, and quality. I will initially try to use weight of the apple as the predictor value.

I eat an apple everyday so I want to investigate what characteristics are significant when looking for an apple.

To analyze the quantitative data, I made a boxplot. Most of the medians are near zero with Crunchiness median closer to 1. Every variable has some outliers. I will look more into them during singular regression. Size, Juiciness, and Acidity have long whiskers (Q1 and Q3). Weight has a lot of outliers especially near -5 and -6. I considered using histogram also but found it is not applicable.

The Kaggle apple quality data set features a unique id for each apple. Each apple has a range value for its characteristics. The apple characteristics are:

- Size of that apple
- Weight of that apple
- Degree of sweetness
- Texture indicating the crunchiness
- Level of juiciness
- Stage of ripeness
- Acidity level
- Quality of that apple

All the characteristics are quantitative, except for the quality of apple. The data contains negative data points because the data set is scaled and falls above/below the mean value to have both positive and negative range values.

The original data set had 4001 rows. I removed almost 3000 rows so I would be able to run computations on the data easily. The categorical Quality values will be converted from text bad/good to numeric 0/1. Also, removed column “A_id” listing the unique id for each apple to do computations.

For model building, I first plotted and found the summary statistics for each of the four variables (Size, Sweetness, Ripeness, Juiciness vs Weight). All the plots had randomness and very low p-values which made each variable statistically significant (which requires them to be addressed in the model).

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Some variables were eliminated:

- Juiciness in the partial significance model was not statistically significant to a linear model that already includes size, sweetness, and crunchiness.
- Crunchiness in the partial significance model was not statistically significant to a linear model that already includes size and sweetness.
- Acidity is important in the limited activity of cooking with apples. I am interested in raw apples, so I did not square it.

The maximum model created included the remaining quadratic terms Size, Sweetness, and Ripeness. The summary showed the polynomial terms were statistically significant given that the original terms were included.

Singular regression graphs were used to identify independent variables. The graphs show randomness throughout.

- Size vs Weight has a few outliers near the -6.5 lb and 5 lb y-axis. Will investigate to see if any of the points are influential. With a t = -5.216 and p = 2.2e-7, reject H_0 and conclude there is a statistically significant linear relationship between Size and Weight.
- The Sweetness vs Weight graph has a few outliers near (-6, -2.5), (5, -2.5), and (5,-1.25). With a t = -4.411, p = 1.13e-5, reject H_0 and conclude there is a statistically significant linear relationship between Weight and Sweetness.
- Ripeness vs Weight has a lot of randomness in the plot and is very spread out. Will want to investigate the (-6, 0), (-4, 5), (-2, -6.5), (6, -1.25) areas. Based on a t = -7.514 and p = 1.21e-13, reject H_0 and conclude there is a statistically significant linear relationship between Ripeness and Weight.

I tried several model transformations. The Box-Cox transformation worked the best. I shifted the y-value Weight by adding seven to each point so all the y-values would be positive to work for the Box-Cox transformation. Then I plotted the best model that resulted in a lambda = 1.07. There were some influential points that could be explored:

- Removed five outliers: 84, 1023, 526, 1063, and 560.
- The studentized residual and residual vs fitted plots were very spread out and random. For the first plot, the fitted value between the residuals y-axis -4 and -6 looked like the observation could be extremely high compared to its prediction scaling for the standard deviation and leverage.
- After removing the outlier and checking the assumptions, there was another influential point near the residuals y-axis -4.
- I repeated that until it seemed like the points were not as widespread.

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- For homoscedasticity, the plot is random, but it is a low p-value (<2.2e-16). Thus, homoscedasticity fails. Normality assumptions are satisfied since the QQ-plot mostly follows the 45-degree line and a high p-value (0.1066).
- Lambda is 0.8282 which is the optimal value of lambda. However, 0.8282 is inside of the 95% CI, so it is a reasonable choice.

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The analysis to identify significant variables was based on the Box–Cox transformation which was applied to stabilize variance and improve linearity when modeling the shifted apple weight. The final selected model included main effects and several interaction terms. Based on the summary output, the following predictors were statistically significant at alpha = .05:

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Main effects: size, sweetness, ripeness, acidity, and the indicator variable for *quality* = *good*.

Quadratic terms: size^2, sweetness^2, and ripeness^2.

Interaction terms:

- size × sweetness
- size × juiciness
- size × acidity
- size × quality
- sweetness × juiciness
- sweetness × ripeness
- crunchiness × juiciness
- crunchiness × ripeness
- crunchiness × quality
- juiciness × acidity
- juiciness × quality
- ripeness × quality

These significant terms indicate that both the quadratic terms and the interactive relationships between variables play an important role in predicting shifted apple weight.

R^2 = 0.3959, 39.59% of variation in shifted apple weight can be explained by a linear relationship of the predictors included in the final model above.

Since this is a predictive model, the professor recommended eliminating any hypothesis testing because the significant variables were already identified. For validation, I did one interpretation of $\hat{\beta}_0$:

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- $\hat{\beta}_0 = 6.2834$, represents the expected shifted weight when all predictors are equal to zero (size = sweetness = ripeness = acidity = 0, and quality = "bad"). Because these values are outside the realistic range for apples, the intercept does not have a meaningful physical interpretation, but it is required to center the regression plane. The intercept is statistically different from zero (t = 57.814, p-value < $2e^{-16}$).

The overall fit of the model evaluates how closely the predicted values match the observed data. Based on the statistical summary of the final model, the F-statistic is 27.35 with a p-value < $2e^{-16}$ so we reject H_0. This indicates that the model as a whole is statistically significant, meaning that at least one of the predictors/quadratics/interactive terms (size, sweetness, ripeness, acidity, or quality good) significantly contributes to explaining variation in shifted apple weight.

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The Root Mean Square Error (RMSE) for the final Box–Cox regression model is 1.1760, which means that, on average, the model's predictions differ from the actual shifted apple weights by about 1.176 units on the shifted scale. A lower RMSE would indicate more accurate predictions, but this is close to the potential limit 0.

The predictive power of a model measures how well its predictions align with actual outcomes for new, unseen data.

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Using LASSO regularization, we assessed how well the model predicts the weight shifted from seven predictors: size, sweetness, crunchiness, juiciness, ripeness, acidity, and quality.

Using 5-fold cross-validation, the optimal LASSO penalty (λ = 0.0011918) minimizes the mean squared prediction error (MSPE) to 1.804, indicating the model's average squared deviation when predicting new data. For comparison, Ridge regression achieves an MSPE of 1.822, so this model's LASSO slightly outperforms Ridge in out-of-sample predictive accuracy.

Elastic Net, tested across alpha values from Ridge = 0 to LASSO = 1, shows similar positive predictive performance. The lowest cross-validated MSPE (1.8127) occurs at alpha = 1, confirming that LASSO provides the best predictive performance among these models.

LASSO also shrinks some coefficients close to zero, which highlights the most relevant predictors while reducing model complexity. Overall, all three models show similar predictive accuracy, suggesting that multicollinearity among predictors is not extreme and that variable selection via LASSO does not reduce predictive power.

The key predictors of apple weight shifted are ripeness, sweetness, size, and quality, as indicated by the largest LASSO coefficients. Focusing on these traits through cultivation

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practices, harvest timing, and quality control can meaningfully increase apple weight. Variables like acidity, juiciness, and crunchiness have minimal impact and can be deprioritized, saving resources. The model's low cross-validated MSPE (1.81) shows it reliably predicts new data, supporting decisions in packaging, storage, and production planning.

Based on the linear regression analysis, a strength for the model is that it can improve customer decision making when picking out an apple. A constraint is the y-value (weight shifted) interpretation limit. I did not do any hypothesis tests since the significant variables were in the model, and the non-significant variables were removed.

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A limitation of the study was that the data was set up as a range from the mean. The y-value (weight) had positive and negative values. I needed to shift the weight, so it was all positive values to perform the Box-Cox transformation. Also, the homoscedasticity assumption is not met since it has a low p-value (<2.2e-16). These limitations only shift the predictive model without any change impact.

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A potential follow up is to use Spline regression which may improve the analysis since it is a regression technique where the response variable is fit to a singular numeric explanatory variable. By using the degree of the spline and number of internal knots, the model can flexibly fit complex patterns in the data that linear regression or LASSO may not capture. This could lead to improved predictive accuracy and a better understanding of how variables like ripeness, sweetness, size, and quality influence shifted apple weight.

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Using the LASSO method, ripeness, sweetness, size, and quality were identified as the key predictors of shifted apple weight. This insight can improve quality control and assurance by focusing on the most influential factors while reducing model complexity. LASSO enhances predictive accuracy and supports data-driven decision-making, helping to increase efficiency and reduce waste in apple production.

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In summary, the variables that are significant to desirability and the variables that are not significant are identified. Producers and consumers can concentrate on what matters using this model. This model can help lead to a healthier diet and longer life.

In the future, I would like to look at additional samples from different locations in the US to see if variables are statistically significant across regions. This would allow me to assess whether the LASSO key predictors of shifted apple weight—ripeness, sweetness, size, and quality—remain statistically significant across diverse growing conditions. I would then reapply the Box-Cox transformation to assess whether quadratic and interaction terms continue to contribute significantly to the model.

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